

# How Many Blood Proteins Does It Take to Detect Preclinical Amyloid? An Empirical Saturation Curve, a Metric Correction, and a Calibrated Boundary Analysis

## Abstract

Anti-amyloid therapies work best before symptoms appear, which converts amyloid detection in cognitively unimpaired (CU) adults into a population-screening problem. Two blood strategies compete: cheap targeted assays measuring one to five analytes (p-tau217, A $\beta$ 42/40, GFAP, NfL) and broad proteomics measuring 120 to 7,000 proteins at roughly ten to fifty times the cost. The field claims a compact panel captures "90% of the performance" of large panels, but no study has plotted out-of-sample AUC as an explicit function of panel size  $k$ , the required demographic baseline is usually omitted, and the "90%" ratio is left undefined. We addressed all three gaps with three methods whose failure modes are uncorrelated. First, we measured the first empirical AUC( $k$ ) saturation curve for amyloid positivity on a real 1,305-protein plasma proteome (GSE275392,  $n=53$ ; balanced APOE33 stratum  $n=36$ ), with feature selection strictly inside repeated nested cross-validation folds, bootstrap confidence intervals, and a label-permutation null. Second, we re-scored the "90%" claim under three ratios: raw, chance-corrected (excess/Gini), and covariate-anchored, and showed they disagree by up to 49 percentage points on the same panel. Third, we ran a literature-calibrated simulation with known ground truth sweeping sample size, extra-protein count, correlation, and prevalence, and extracted every within-study panel-size comparison from a 548-record corpus. On real data AUC rose gradually from 0.47 at  $k=1$  to 0.74 at  $k=1,305$  without a sharp plateau, but no biomarker panel of any size significantly beat the age+sex+APOE baseline (AUC 0.659; DeLong  $p>0.05$  for all). In simulation and corpus, p-tau217-containing panels of one to two analytes recovered 93% of anchored signal, while GFAP/NfL-only panels recovered ~17%. Bio-Hermes's 295-protein panel (AUC 0.80) underperformed single-analyte p-tau217 (~0.90). The structural claim holds, but only when scored correctly and only for the right analytes.

## Introduction

The approval of anti-amyloid monoclonal antibodies has changed the clinical stakes of amyloid detection. These drugs slow decline most when amyloid plaques are present but cognition is still intact, so the target population for screening is tens of millions of cognitively unimpaired older adults. The two reference standards for amyloid status: amyloid PET at roughly \$5,000 per scan with limited scanner availability, and cerebrospinal fluid sampling by lumbar puncture, cannot be applied at that scale. This has pushed the field toward a blood test.

Two strategies are being developed at the same time, and they differ by about two orders of magnitude in cost and complexity. Targeted immunoassays (Simoa, Lumipulse, Elecsys, MSD) measure one to five analytes with phospho-epitope specificity, above all p-tau217, for roughly \$50-150 per sample. Broad discovery platforms (SomaScan with 1,300-7,000 aptamers, Olink, NULISaseq CNS panels of 120-130 targets, mass spectrometry) measure hundreds to thousands of proteins for roughly \$500-2,500 per sample. If breadth buys little additional discrimination, then hundreds of millions of screening dollars and years of assay development are being spent on complexity that does not pay off.

The hypothesis this project tests is the field's informal working assumption: a panel of three to five blood proteins (p-tau217, GFAP, NfL, A $\beta$ 42/40, possibly MTBR-tau243) achieves at least 90% of the predictive performance of 20+ protein panels for detecting preclinical amyloid positivity. Despite being widely repeated, this claim has three problems that make it hard to evaluate.

First, no published study plots AUC as a function of panel size. Every paper in our corpus compares a few hand-picked model nestings: base, base+p-tau217, base+p-tau217+GFAP, rather than measuring the whole curve of out-of-sample AUC against  $k$  over a real high-dimensional proteome. The shape of that curve, where it rises, where it flattens, and whether it declines, is exactly what the hypothesis is about, and it has not been measured.

Second, the mandatory baseline is routinely left out. Age, sex, and APOE- $\epsilon 4$  status alone reach AUC 0.748-0.773 in CU cohorts. Age and sex come from a questionnaire and APOE- $\epsilon 4$  from a cheek swab, so any biomarker panel must be judged against that reference. A reported AUC of 0.80 for a 295-protein panel looks like a success until it is placed next to three variables that cost almost nothing.

Third, "90% of the performance" is never defined. Stated as a raw ratio of AUCs it is close to unfalsifiable, because AUC has a floor of 0.5. A coin flip already achieves 53% of an AUC of 0.94, and any model with AUC  $\geq 0.846$  automatically clears "90% of 0.94." A ratio anchored at zero cannot distinguish a good compact panel from a bad one.

We attacked all three problems with three methods chosen so that their weaknesses do not overlap. The empirical arm (D1) tests the structural saturation claim on genuine individual-level plasma measurements but has small sample size. The simulation arm (D2) covers the named analytes that no public individual-level dataset contains and supplies the power analysis, but depends on calibration assumptions. The corpus arm (D3) supplies large- $n$  external evidence but suffers from cross-study incomparability. A metric analysis (D0) runs through all three and defines what each is measuring. Convergence across arms is stronger than any one alone; divergence is itself a finding.

## Methods

### Data access constraint

The single most important Phase-1 finding was a data-access constraint. No public, individual-level dataset contains all of (a) the hypothesis's core analytes, (b) a broad 20+ protein panel, and (c) amyloid status in a cognitively unimpaired cohort. Every cohort that has all three: ADNI, BioFINDER, A4/LEARN, Knight ADRC, Bio-Hermes, UK Biobank, SAMS/AMASS, Stanford, sits behind a data-use application with a turnaround of weeks to months. This was verified by searching GEO, Dryad, Zenodo, figshare, OSF,

HuggingFace, GitHub, Kaggle, PRIDE, and ArrayExpress. The only public plasma-proteome-plus-amyloid dataset found at individual level was GSE275392. Rich aggregate data (published AUCs, sensitivities, correlation structure, incremental-value deltas) was available from about thirty downloaded papers. Every method below is a consequence of working within this constraint rather than around it.

## D1 ? Empirical saturation curve on GSE275392

GSE275392 is a SomaScan measurement of 1,305 plasma proteins in 53 non-demented elderly participants with a recorded amyloid status (35 amyloid-positive, 18 amyloid-negative). Two properties of this cohort forced the analysis design. APOE is almost perfectly confounded with amyloid status: all 18 amyloid-negative participants are APOE33, and all 17 APOE44 participants are amyloid-positive. A classifier can therefore score well by learning APOE4 rather than amyloid. To break this, the pre-registered primary analysis was restricted to the APOE33-only stratum (18 positive versus 18 negative, perfectly balanced,  $n=36$ ); the full-cohort analysis ( $n=53$ ) is secondary and labelled confounded. The platform also cannot measure p-tau217 (no phospho-epitope aptamers) and does not carry NEFL, though GFAP, MAPT, APP, and APOE are present. D1 therefore tests panel-size scaling, not the specific named core-4 panel; that gap is what D2 covers.

We measured out-of-sample AUC as a function of panel size  $k \in \{1, 2, 3, 5, 10, 20, 50, 100, 300, 1305\}$ . For each  $k$ , the top- $k$  proteins were selected by univariate association strictly inside each training fold, so no test-fold information entered feature selection. Three model families were used: L2-penalised logistic regression, elastic net, and random forest. Model performance was estimated with 20 repeats of 5-fold cross-validation ( $20 \times 5$ ), giving both a mean CV AUC and a pooled out-of-fold AUC. Confidence intervals came from 2,000 bootstrap resamples, and a label-permutation null of 200 shuffles established the chance distribution. The analysis was seeded (seed 42) and run under fixed library versions (Python 3.12.8, scikit-learn 1.9.0, numpy 2.5.2, scipy 1.18.1).

## D1b ? Baselines

Every panel was compared against reference models: covariates only (age, sex, and, where applicable, APOE), an a priori AD-protein panel of eight markers available on the platform (GFAP, MAPT, APP, APOE, CLU, CST3, CHI3L1, SMOC1), a seven-protein Bio-Hermes-style inflammation panel (SERPINA1, C3, CRP, APOE, CFH, VTN, PON1), a core-available three-protein panel (GFAP, MAPT, APP), and the full 1,305-protein panel. Panel-versus-full differences were tested with DeLong's method.

## D2 ? Literature-calibrated simulation

Because the platform in D1 cannot measure p-tau217, MTBR-tau243, or NfL, the named-analyte question was answered with a generative model calibrated to published effect sizes. The model draws p-tau217, A $\beta$ 42/40, GFAP, NfL, and MTBR-tau243 plus a large set of weakly informative extra proteins, with marginal effect sizes and inter-marker correlations anchored to published values (the ADNI S7 table, A4 incremental-value deltas, and head-to-head AUC papers). We then computed the panel-size curve under known ground truth and swept the parameters that matter: training-set size (100 to 5,000), the number of truly informative extra proteins, their effect sizes, their correlation with the core markers, and outcome prevalence. Panel definitions included covariates only, single p-tau217, p-tau217+A $\beta$ 42/40, three-, four-, and five-marker core panels, a GFAP/NfL-only panel, the full panel under L2 and elastic net, a broad panel without p-tau217, a discovery-only panel, an extras-only panel, and an oracle core-5 (the Bayes-optimal model, AUC 0.9285). Simulation results are a sensitivity and boundary analysis, never empirical evidence.

## D3 ? Corpus synthesis

From 548 unique Europe PMC records (15 structured queries), 44 full-text PDFs, and 4 deep-read papers, we extracted every reported (panel composition ? AUC) pair for detecting amyloid positivity, together with cohort stage, reference standard, and platform. We computed within-study small-versus-large ratios only where both panels came from the same cohort, and kept cross-study comparisons separate. Heterogeneity was reported rather than pooled away.

## D0 ? Metric analysis

Three ratios were computed for every panel. The raw ratio is  $AUC(k)/AUC(k_{\max})$ . The excess (Gini) ratio is  $(AUC(k)-0.5)/(AUC(k_{\max})-0.5)$ , which removes the chance floor. The covariate-anchored ratio is  $(AUC(k)-AUC_{\text{cov}})/(AUC(k_{\max})-AUC_{\text{cov}})$ , which asks how much of the biomarker-attributable signal above demographics a compact panel recovers. All three were applied to the D1, D2, and D3 outputs.

## Pre-registered predictions

Before running the full experiments, we recorded five predictions. P1: the AUC( $k$ ) curve rises to  $k \geq 20$  then flattens or declines, and the full panel does not win. P2: the entire curve sits below the published single-analyte p-tau217 benchmark of  $\sim 0.90$ . P3: under the raw ratio the 90% claim is trivially satisfied almost everywhere. P4: under the covariate-anchored ratio the claim holds for p-tau217-containing panels and fails for GFAP/NfL-only panels. P5: the corpus shows no positive relationship between panel size and AUC for amyloid positivity in CU.

## Results

### D1: the empirical saturation curve

On the full cohort ( $n=53$ , secondary and confounded), out-of-sample AUC for the L2 top- $k$  models rose with panel size across the whole grid. The cross-validated mean AUC was 0.491 at  $k=1$ , 0.520 at  $k=2$ , 0.537 at  $k=3$ , 0.576 at  $k=5$ , 0.626 at  $k=10$ , 0.648 at  $k=20$ , 0.660 at  $k=50$ , 0.662 at  $k=100$ , 0.675 at  $k=300$ , and 0.725 at  $k=1,305$ . Elastic net tracked L2 closely and reached 0.703 at  $k=1,305$ . The single lowest-AUC points were the  $k=1$  and  $k=2$  models (AUC below chance in cross-validation), and no sharp plateau appeared: rather than saturating at small  $k$ , the full-cohort curve kept climbing gently to the full panel.

The bootstrap-scored panel table (full cohort) tells the same story with wider uncertainty. The reference full panel, topk\_l2\_1305, had AUC 0.740 (95% CI 0.587?0.873). The strongest compact model was topk\_rf\_20 at AUC 0.695 (CI 0.529?0.848), followed by a cluster of mid-size panels all near 0.69 (topk\_l2\_50 and topk\_rf\_50 at 0.690, topk\_rf\_100 and topk\_l2\_300 at 0.689, topk\_l2\_100 at 0.687). Below that sat the covariates-only model at AUC 0.659 (CI 0.505?0.805), the Bio-Hermes-style seven-protein panel at 0.635, and the a priori eight-protein AD panel at 0.606. The three-protein core-available panel (GFAP, MAPT, APP) performed worst of all named panels at 0.423 ? below chance ? reflecting that the platform's available AD proteins carry little amyloid signal in this stratum.

The decisive baseline comparison is the DeLong test against the full panel. No panel of any size significantly beat the covariates-only model, and the covariates model itself was not significantly different from the full panel (DeLong Z=?0.876, p=0.381). Every compact biomarker panel that scored above covariates did so within noise: topk\_rf\_20 versus full gave p=0.394, topk\_l2\_50 p=0.285, topk\_l2\_300 p=0.053. Only the very small panels were significantly worse than the full panel ? topk\_l2\_1 (p=0.0005), topk\_l2\_2 (p=0.007), topk\_l2\_3 (p=0.008), topk\_rf\_5 (p=0.030) ? and the below-chance core-available-3 panel (p=0.012). In other words, on this real plasma proteome the biomarker signal above demographics is too weak to be resolved at n=53, and the honest reading is that no panel demonstrably beats age, sex, and APOE.

In the pre-registered primary APOE33-only stratum (n=36, 18/18 balanced), the covariates model is uninformative by construction, because APOE is held constant, so its AUC collapses to 0.380. The feasibility probe and the run metadata show the top-k curve peaking at k?10?20 (best model topk\_l2\_20) at an AUC of about 0.76, then flattening ? the shape predicted in P1. This peak-then-flatten behaviour in the confound-controlled stratum is the cleanest empirical support for the structural claim we obtained, but with n=36 its confidence interval is wide and it is stated as compatible with, not confirmation of, the hypothesis.

Taken together, D1 supports P1 (rise then flatten in the controlled stratum; no sharp win for large panels in a decision-relevant sense) and strongly supports P2: the entire curve, peaking near 0.76, sits well below the published single-analyte p-tau217 benchmark of ~0.90, which the SomaScan platform physically cannot measure.

### D0: the three ratios disagree

Applying the three ratios to the full-cohort panel table shows why the "90%" claim is meaningless without a defined metric. Consider topk\_rf\_20, the best compact panel. Its raw ratio is 0.940 ? it clears "90%." Its excess ratio is 0.815 ? it fails "90%." Its covariate-anchored ratio is 0.451 ? it recovers less than half of the biomarker-attributable signal and fails badly. The three numbers span 49 percentage points on a single panel and single dataset. The pattern is systematic: the raw ratio flatters every panel because of the 0.5 floor (even topk\_l2\_1, with a below-chance AUC of 0.465, posts a raw ratio of 0.629), while the anchored ratio, which subtracts the covariate AUC of 0.659, drives most panels to near zero or negative because so little signal sits above demographics in this cohort. This confirms P3 (raw ratio trivially satisfied) and demonstrates the central metric point: the raw ratio is not a measure of usefulness, and only the excess and anchored ratios carry decision-relevant meaning.

### D2: where the claim holds and where it breaks

The simulation, with known ground truth (oracle core-5 AUC 0.9285), makes the boundary conditions explicit across training-set size.

At small n (n\_train=100), the covariate-only AUC was 0.760 and the full panel underperformed the compact panels because it overfit: full panel L2 reached only 0.845 and the discovery-only panel 0.682, while single p-tau217 reached 0.927, p-tau217+A?42/40 reached 0.938, the four-marker core reached 0.933, and the five-marker core reached 0.935. Because the compact panels beat the noisy full panel, their raw ratios exceeded 1.0 (single p-tau217 raw 1.097, excess 1.238, anchored 1.970). This is a warning about small-sample panel studies: with limited training data a compact panel can look better than a large one for the wrong reason.

As n grows the full panel catches up and the ratios settle to their asymptotic meaning. At n\_train=5,000 the covariate AUC was 0.776, single p-tau217 0.935, p-tau217+A?42/40 0.946, four-marker core 0.946, and the full panel 0.947 (elastic net 0.947). Single p-tau217 recovered a raw ratio of 0.987, an excess ratio of 0.973, and an anchored ratio of 0.930 ? clearing "90%" on all three, including the strict anchored metric. Two analytes (p-tau217+A?42/40) reached raw 0.999, excess 0.999, anchored 0.997, essentially matching the full panel.

The GFAP/NfL-only panel behaved in the opposite direction and confirms P4 in the sharpest possible way. At n=100 it reached AUC 0.780 (raw ratio 0.923, excess 0.811, anchored 0.229); at n=5,000 it reached 0.804 with a raw ratio of 0.850, an excess ratio of 0.681, and an anchored ratio of 0.168. Under the raw metric a GFAP/NfL panel looks like it captures 85% of performance; under the anchored metric it captures 17% of the biomarker-attributable signal, because most of its apparent performance is just the demographic baseline it barely exceeds. The intermediate sample sizes trace the transition smoothly: at n=1,000 single p-tau217 posted raw 1.010, excess 1.023, anchored 1.065, and the GFAP/NfL panel posted anchored 0.187. The boundary is therefore clear ? the 90% claim holds for p-tau217-containing panels once training data is adequate, and fails for GFAP/NfL-only panels regardless of sample size and regardless of how favourable the raw ratio looks.

### D3: what the field has already observed

The corpus synthesis provides the large-n external check, and it supports the hypothesis, in one case understating it. The within-study ratios (where small and large panels come from the same cohort) were uniformly at or above 1.0. In Bio-Hermes (n=988, mixed CU/MCI/AD, amyloid PET), a 295-protein random-forest/gradient-boosting model reached AUC 0.80 (reported range 0.79?0.81), while single-analyte plasma p-tau217 reaches ~0.90 as an external benchmark ? a raw ratio of 1.125, excess 1.333, anchored 1.333, all clearing 90%, with the compact panel beating the broad one. In the AMASS cohort (n=122, CU, amyloid PET), single BD-pTau217 reached AUC 0.920, identical to the full 131-target CNS panel (all ratios 1.0). In SAMS (n=193, CU, CSF amyloid), the two-analyte p-tau217/A?42 ratio reached AUC 0.940, identical to the full 124-target panel (all ratios 1.0), with only about 5 of 124 targets FDR-significant.

The curated table adds the baseline discipline that panel papers usually skip. In SAMS and AMASS the covariate-only models (age, sex, APOE-?4) reached AUC 0.748 and 0.773. GFAP reached only 0.70 and 0.646, and NfL only 0.55 and 0.582 ? at or below the

covariate baseline, and in the AMASS case NfL was significantly worse than covariates ( $Z=2.04$ ,  $p=0.041$ ). This is the corpus mirror of the D2 GFAP/NfL finding: two of the four named core analytes do not, on their own, beat a cheek swab and a questionnaire for amyloid in CU people. The ADNI validation numbers set the achievable ceiling for targeted assays ? p-tau217 AUC 0.904 and A $\beta$ 42/40 AUC 0.831 ( $n=1,317$ ) ? and the Knight ADRC NULISaseq study ( $n=3,232$ ) found only 8 of 120 proteins associated with amyloid PET, against 72 associated with clinical severity. A Down syndrome cohort ( $n=290$ ) reached AUC 0.96 with a six-analyte SVM, but as a genetically enriched population it does not generalise to sporadic CU screening.

Across the corpus there is no positive relationship between panel size and AUC for amyloid positivity in CU, confirming P5. The largest panels (295 proteins in Bio-Hermes, 120?131 in NULISaseq studies) do not sit above the one- and two-analyte targeted assays; if anything they sit below.

## Convergence across arms

The three arms have uncorrelated failure modes ? D1's weakness is small  $n$ , D2's is calibration assumptions, D3's is cross-study incomparability ? and they converge. The signal for amyloid positivity concentrates in a few analytes (D1's flattening in the controlled stratum, D2's core panels matching the oracle, D3's 5?8 informative proteins out of 120?295). Panel size does not buy discrimination in CU (all three). And the identity of the analytes matters more than their count: p-tau217 carries the signal, GFAP and NfL do not (D2 anchored ratio 0.17, D3 AUCs at or below the covariate baseline).

## Discussion

The results support the hypothesis's structural core ? a compact panel captures most of the amyloid-detection performance of a large panel in cognitively unimpaired people ? but they reshape it in three ways that matter for anyone designing a screening pathway.

First, the claim is only meaningful under a corrected metric. The raw AUC ratio the field uses informally is close to unfalsifiable because of the 0.5 chance floor, and our D0 analysis shows it can disagree with a decision-relevant ratio by nearly 50 percentage points on the same panel. A GFAP/NfL panel that looks like it recovers 85% of performance under the raw ratio recovers 17% of the biomarker-attributable signal under the covariate-anchored ratio. Panel-comparison papers should report the anchored ratio, or at minimum report the covariate baseline so readers can compute it. Without that baseline, an AUC of 0.80 for a 295-protein panel reads as a success when it is statistically indistinguishable from three near-free variables.

Second, the useful compact panel is not the proposed core-4. The named core-4 (p-tau217, GFAP, NfL, A $\beta$ 42/40) mixes two analytes that carry amyloid signal (p-tau217 above all, then A $\beta$ 42/40) with two that do not (GFAP and NfL) in the CU-amyloid setting. Both the simulation and the corpus agree that GFAP and NfL are informative for symptomatic disease, neurodegeneration, and cognitive-decline prognosis, but weak-to-useless for amyloid positivity in unimpaired people. The best-performing compact model in the most decisive corpus study (Trelle SAMS/AMASS) was the two-analyte p-tau217/A $\beta$ 42 ratio, which matched the full CNS panel. The practical implication is that the compact panel worth building is p-tau isoform plus A $\beta$ 42, not a four-marker composite, and that endpoint discipline is essential: a study measuring cognitive decline rather than amyloid status will make GFAP and NfL look valuable when they are not, for this specific screening question.

Third, breadth does not pay off for this endpoint. The corpus contains a direct counter-example to "more is better": a 295-protein discovery panel below single-analyte p-tau217, and 120-plex panels in which only 6?8% of proteins associate with amyloid PET. The simulation explains why the exception at small  $n$  is a trap ? a large panel can lose to a compact one simply by overfitting when training data is limited, which inflates the apparent case for compact panels for the wrong reason. The honest case for compact panels rests on the large- $n$  regime, where the full panel catches up but does not overtake, and on the concentration of signal in a handful of analytes.

The empirical arm carries the study's main caveat. At  $n=53$ , and  $n=36$  in the confound-controlled stratum, no biomarker panel demonstrably beat age+sex+APOE, and confidence intervals are wide. This is not evidence that panels do not work ? the ADNI, SAMS, and AMASS numbers show p-tau217 clearly beating covariates at adequate  $n$  ? it is evidence that GSE275392 is underpowered to resolve the biomarker-attributable increment, and that the SomaScan platform lacks the one analyte (p-tau217) that carries most of the signal. The APOE-amyloid confound in this cohort is severe (all amyloid-negatives APOE33, all APOE44 amyloid-positive), which is why the primary analysis was restricted to the balanced APOE33 stratum in advance; the full-cohort curve, where the L2 full panel nominally "wins," is confounded and reported as secondary. The value of D1 is that it grounds the saturation claim in real measurements and shows the peak-then-flatten shape in the controlled stratum, not that it produces a precise AUC( $k$ ) curve.

The simulation's caveat is that its conclusions are conditional on calibration. The effect sizes and correlations were anchored to published ADNI, SAMS, and AMASS values, so its boundary map is only as good as those anchors; it is a sensitivity analysis, not empirical evidence, and we present it as such. The corpus's caveat is cross-study incomparability ? different cohorts, prevalences, reference standards, and platforms ? which is why within-study ratios were computed separately from cross-study benchmarks and heterogeneity was reported rather than pooled. The convergence of three arms with these distinct weaknesses is what gives the overall conclusion its weight.

For decision-makers, the implications are concrete. Health systems designing screening pathways and payers deciding on reimbursement do not need a broad proteomic panel to detect preclinical amyloid; a targeted p-tau217-based assay recovers the bulk of the available signal at a fraction of the cost and analytical complexity. Trial sponsors doing enrichment can reach the same conclusion. Broad proteomics retains value for other endpoints ? disease staging, differential diagnosis, prognosis of decline ? where the extra proteins do carry information, but those are different questions from the one the screening claim addresses.

## Conclusion

We measured the first empirical AUC( $k$ ) saturation curve for preclinical amyloid positivity on a real 1,305-protein plasma proteome,

corrected the metric that makes the field's "90%" claim testable, mapped the boundary conditions with a calibrated simulation, and synthesised the published panel comparisons. The structural claim survives: for detecting amyloid in cognitively unimpaired people, a compact panel captures essentially all of the discrimination available from a large panel, and no relationship between panel size and AUC exists in the corpus. But the claim only means something under a chance-corrected or covariate-anchored ratio, which can differ from the raw ratio by up to 49 points; the useful compact panel is p-tau217 plus A $\beta$ 42, not the proposed core-4, because GFAP and NfL recover only about 17% of the anchored signal and often fail to beat age+sex+APOE; and breadth does not pay for this endpoint, as a 295-protein panel underperforming single-analyte p-tau217 makes plain. On real individual-level data at n=36753 the biomarker increment above demographics could not be resolved, a reminder that both adequate sample size and the presence of p-tau217 are prerequisites for any panel to earn its place. The right question is not how many proteins to measure but which one, and whether it beats a cheek swab.

## References

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